Guerroudj Abdennour

Artificial Intelligence Engineering Student

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Professional Summary

Fifth-year AI Engineering student specializing in AI, LLMs, multi-agent systems, and data science, with experience in RAG, fine-tuning, and agentic automation. Proven ability to build advanced AI systems with real-world impact.

EDUCATION

Artificial Intelligence Engineer
National Higher School of Artificial Intelligence 2022 - 2027 (5th Year)

Core Curriculum:
Machine learning and deep learning • Data mining and analysis • Mathematical statistics • Computer vision and NLP • Networks and protocols • System and network security • Operating systems • Software engineering • Databases and data warehouses • Time series and forecasting • Business management and marketing • Entrepreneurship and innovation • Operations research

TECHNICAL EXPERTISE

AI & Machine Learning
- Deep Learning & Natural Language Processing
- Large Language Models (LLMs), RAG & Fine-Tuning
- Multi-Agent Systems & Agentic AI
- AI Workflow Automation (n8n, Autonomous Pipelines)
- Computer Vision Applications
- Data Science & Advanced Analytics

Software Development
- **Mobile:** Android (Kotlin, Java), React Native, Flutter
- **Web:** Full-Stack, React, Next.js, Django, Flask
- **Databases:** PostgreSQL, MongoDB, MySQL

Networks & Security
- Network Architecture & Protocol Design
- TCP/IP Suite & Routing Protocols
- Cybersecurity & Vulnerability Assessment

CREATIVE MEDIA

- Video Editing
- Animation
- Motion Graphics

LANGUAGES

Arabic • French • English

COMPETITIONS & EXPERIENCES

AI Competitions
- **BMS Electric Hackathon**
  Developed an AI application for data warehouse management to improve data organization, processing, and decision support.
- **Algerie Telecom Hackathon**
  Built an AI-based solution for optimizing maintenance and support.
- **Ramy AI Innovation Hackathon**
  Built an AI sentiment analysis system for customer feedback and brand monitoring.
- **ICOSNET Professional Datathon – AI & Data Analysis**
  Professional datathon solving 8 advanced data analysis problems in ML, statistical modeling, and large-scale analytics.

PROFESSIONAL EXPERIENCE

- **Data Science Intern – SONATRACH**
  Performed seismic data analysis and early research on deep-learning fault detection workflows.
- **Research Intern (6 Months) – Institut FEMTO-ST, France**
  6-month research internship at Institut FEMTO-ST — one of the largest scientific research institutes in France in engineering sciences and information and communication technologies. Focused on applied AI and information technologies.
- **AI Engineer – Algerian National People's Assembly (APN)**
  Designed and deployed a RAG system for legislative document search; built automated speech-to-text and summarization to enhance accessibility of sessions and archives.
- **AI Research Engineer – Ministry of Higher Education and Scientific Research**
  Built a large-scale RAG platform for national research documentation access and semantic search.
- **Software Engineer – Algérie Poste**
  Developed a scalable multi-store regional application with tracking, monitoring, and secure data handling.
- **AI Engineer – Voxtra**
  Built Voxtra, an Arabic speech transcription platform with LLM-based multi-level summarization for structured hierarchical summaries.
- **AI Engineer – Multi-Agent Visual Collaboration Platform – Finland Startup Program**
  Engineering a Miro-class multi-agent collaborative workspace with autonomous agents for real-time ideation, visual planning, and workflow orchestration; selected for the Finland Startup Program.
- **AI Engineer – TEXA: Multi-Agent Tourism Intelligence Platform**
  Architected a multi-agent AI platform for tourism enhancement, predictive demand modeling, itinerary optimization, and intelligent exploration and trip planning.

Ongoing Researches

- **Symbolic Latent Space for Optimized LLM Language**
  Pioneering research on symbolic, interpretable latent representations to optimize LLM language efficiency, compositional reasoning, and low-cost inference.
- **AI-Driven Deepfake Detection Research – University of Montreal**
  Advanced research on AI-driven deepfake detection, building deep learning models for facial manipulation, synthetic speech, and multimodal forgery detection to ensure media authenticity and digital trust.
- **SAC-GRAPH5G**: **Soft Actor-Critic controlled Residual GraphSAGE** for **5G wireless network optimization** using transductive k-NN graph construction and dynamic hyperparameter search.